

QC II - Part 2: Application

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1 Electron Correlation

1.1 Multireference Methods

Electron correlation is a fundamental concept in quantum chemistry, referring to the interactions between electrons that are not captured by mean-field approaches such as Hartree-Fock (HF). Accurate treatment of electron correlation is essential for reliable predictions of molecular properties, especially when the electronic structure cannot be adequately described by a single determinant. This distinction gives rise to single-reference and multireference methods.

The dissociation of H-F was studied using different methods for H-F bond distances between 0.5 Å and 4.0 Å using the def2-TVZP^[1] basis set. The calculation was done using ORCA 5.0.4.^[2,3] The following methods were used: restricted Hartree-Fock (RHF), unrestricted Hartree-Fock (UHF), Møller-Plesset perturbation theory of second order (MP2),^[4] coupled cluster with single, double and perturbative triple excitations (CCSD(T)), CASSCF(2,2) and CASSCF(6,6). Complete active space self-consistent field (CASSCF)^[5] is a multireference method that was used in two different ways, first with two active electrons in two active orbitals (CASSCF(2,2)) and second with six active electrons in six active orbitals (CASSCF(6,6)).

Figure 1 shows the dissociation curves for the different methods. The calculated energies are calculated relative to the dissociation limit of the H-F molecule. The dissociation limit is defined as the energy of a hydrogen atom and a fluorine atom at infinite distance.

The RHF and UHF methods show a dissociation curve that is not suitable for describing the dissociation of H-F. The RHF curve approaches relative energies larger than 0 kcal/mol at the dissociation limit, which is not physically meaningful. This is due to the problem that static electron correlation is not accounted for, as each pair of electrons needs to occupy the same spatial wave function. As a result of this, the ionic dissociation of H-F is not described correctly, as the open-shell character can not be represented.

The problem of static electron correlation is absent in the UHF curve, showing the correct dissociation limit, because UHF partially captures static electron correlation by spin symmetry breaking.

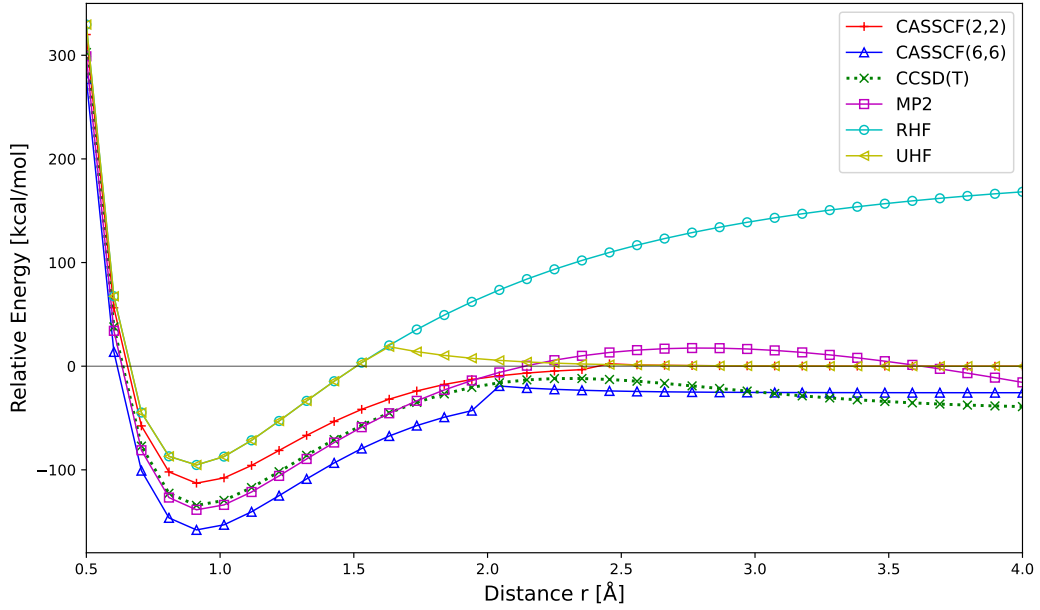


Figure 1: H-F Dissociation for various methods.

In order to obtain the most optimal result for the UHF curve, the calculation was executed within the range of 0.5 Å to 4.0 Å and from 4.0 Å to 0.5 Å. The lowest energy value was obtained for each distance, as instructed by the tutor to minimize the software error. The software ORCA uses the previous solution for the calculations, which results in a difference in the symmetry breaking depending on the direction of the calculation.

Nevertheless, the UHF curve shows a unphysical behaviour at 1.6 Å, where the relative energy has a maximum. This is due to the fact that the UHF method allows for spin symmetry breaking, which can lead to unphysical results. Furthermore, both RHF and UHF methods calculate a binding energy at the equilibrium that is too high due to missing dynamic electron correlation.

Post HF methods like MP2 and CCSD(T) show a much better description of the dissociation of H-F, because they take the dynamic correlation effects into account. The MP2 energy correction is calculated as follows:

$$E_0^{(2)} = \frac{1}{4} \sum_{i,j,ab} \frac{|\langle ij||ab \rangle|^2}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b}, \quad (1)$$

where $\langle ij||ab \rangle$ are the two-electron integrals and ϵ_i , ϵ_j , ϵ_a and ϵ_b are the orbital energies of the occupied and virtual orbitals.

However, this results that the MP2 energy correction approaches $-\infty$ for large distances, which is not physically meaningful. This occurs because the energy difference between occupied and virtual orbitals decreases for large distances, which leads to denominator approaching zero. A similar argument can be made to a lesser extent regarding CCSD(T), given the perturbative calculation of triple excitations. Nevertheless, both MP2 and CCSD(T) show a correct and similar behavior at the equilibrium, as both account for dynamic electron correlation.

In total, single reference methods like RHF, UHF, MP2 and CCSD(T) do not take static electron correlation into account, which is necessary for a correct description of the dissociation of H-F. Furthermore, some single reference methods like MP2 and CCSD(T) include dynamic electron correlation, which leads to a better description of the dissociation of H-F.

Both CASSCF methods describe the dissociation of H-F correctly as they are multi-reference methods, which explicitly handle static electron correlation via the active space. Nevertheless, the CASSCF methods have an unphysical behavior at 2.5 Å for the CASSCF(2,2) and at 2.0 Å for the CASSCF(6,6) method, which is not as drastic as in the UHF curve. The CASSCF(2,2) method computed to high energies at the equilibrium distance, but the description of the dissociation limit is accurate with only 2 active electrons in 2 active orbitals. The CASSCF(6,6) method shows a better description of the equilibrium distance and the dissociation limit.

In summary, the CASSCF method includes static electron correlation, which allows the description of the H-F dissociation, but does not describe dynamic electron correlation well. Further improvements can be made by using post-CASSCF methods like the complete active space with perturbation theory of second order (CASPT2) method, which is a post-CASSCF method that accounts for dynamic electron correlation.

1.2 Carbenes

The calculation of the singlet-triplet energy splitting ΔE_{S-T} for two carbenes, namely Methylene and *p*-Benzyne, and the Methylene H-C-H angles α_{HCH} were computed using the def2-TZVP^[1] basis set and the following methods: HF, MP2,^[4] CCSD(T), Becke 3-parameter exchange functional with Lee-Yang-Parr correlation functional (B3LYP),^[6,7] Tao-Perdew-Staroverov-Scuseria (TPSS)^[8] and Perdew-Wang 6-parameter exchange functional with Becke-95 correlation (PW6B95).^[9] The geometries for the CCSD(T) calculations were optimized using the MP2 method in a def2-TZVP basis set. The structures of the carbenes are shown in Figure 2.

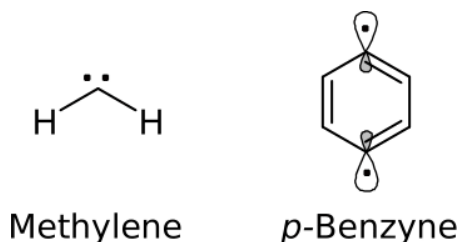


Figure 2: Structures of Methylene (left) and *p*-benzyne (right). Figure taken from reference [10].

The calculations for the geometry optimizations and the single point calculations were done using TURBOMOLE V7.7.1.^[11] The results for the singlet-triplet energy splitting ΔE_{S-T} and the Methylene H-C-H angles α_{HCH} are shown in Table 1.

Table 1: Singlet-Triplet energy splittings (kcal/mol) for *p*-benzyne and methylene along methylene H-C-H angles. All values were obtained using a def2-TZVP basis. CCSD(T) calculations were performed using MP2 geometries.

Method	ΔE_{S-T} [kcal/mol]		α_{HCH}	
	<i>p</i> -Benzyne	Methylene	Singlet	Triplet
HF	80.4	28.4	103.7°	131.8°
MP2	-28.6	15.8	102.4°	133.1°
CCSD(T)	-3.3	10.8	-	-
B3LYP	13.4	11.9	101.9°	135.1°
TPSS	7.0	17.8	101.3°	135.4°
PW6B95	14.3	12.6	101.6°	134.8°
Experiment	-4.2	9.0	102.4°	135.5°

The only method that describes the singlet-triplet energy splitting ΔE_{S-T} with almost chemical accuracy (1 kcal/mol) for both systems is CCSD(T). The other methods show larger deviations from the experimental values. HF overestimates the singlet-triplet energy splitting for both systems and results in too large ΔE_{S-T} values. This is due to the fact that a triplet needs to be calculated using UHF, which contains spin contamination and therefore lowers the energy.

MP2 underestimates the ΔE_{S-T} for p-benzyne and overestimates the ΔE_{S-T} for methylene. Nevertheless, MP2 is the only method beside CCSD(T) that describes the singlet-triplet energy splitting for p-benzyne with a negative value, which is in line with the experimental value and also gives a reasonable value for the ΔE_{S-T} of methylene. The large error in the p-benzyne ΔE_{S-T} is due to the small HOMO-LUMO gap, which leads to a large energy correction of the correlation energy (see Equation 1). Furthermore, MP2 does not account for the static correlation effects present in p-benzyne.

The different density functional theory (DFT) functionals follow the trend of the *Jacob's Ladder* for the singlet-triplet energy splitting ΔE_{S-T} of methylene. The meta-GGA TPSS is worse than the hybrid functionals B3LYP and PW6B95. The singlet-triplet energy splitting ΔE_{S-T} of p-benzyne does not exactly follow this trend, because the meta-GGA TPSS shows a smaller deviation of the ΔE_{S-T} than the two hybrid functionals. Nevertheless, it is notable that the two hybrid functionals give similar singlet-triplet energy splitting ΔE_{S-T} for both systems.

The H-C-H angles for methylene are also affected by the choice of method, but all methods predict a similar results for the angle and are in good agreement with the experimental values. It is to be noted that the angles for HF again have the largest deviations from the experimental values.

The CCSD(T) method is recommended for very precise calculations of the singlet-triplet energy splitting ΔE_{S-T} and the H-C-H angles for both carbenes. For the singlet-triplet energy splitting ΔE_{S-T} of p-benzyne, no other method is recommended. The calculation of the singlet-triplet energy splitting ΔE_{S-T} of methylene can be done with CCSD(T) for the most accurate results or with B3LYP or PW6B95 for still reasonable results with reduced scaling. The H-C-H angles can be calculated with all methods beside HF or MP2, as they all give similar results.

2 Basis Set Convergence

The convergence of different basis sets is an important aspect of quantum chemical calculations. In this report, the basis set convergence of the formic acid dimer is investigated. The structure of the formic acid monomer and dimer are shown in Figure 3 and were optimized using the TPSS^[8]-D3^[12]/def2-TZVP^[1] level of theory.

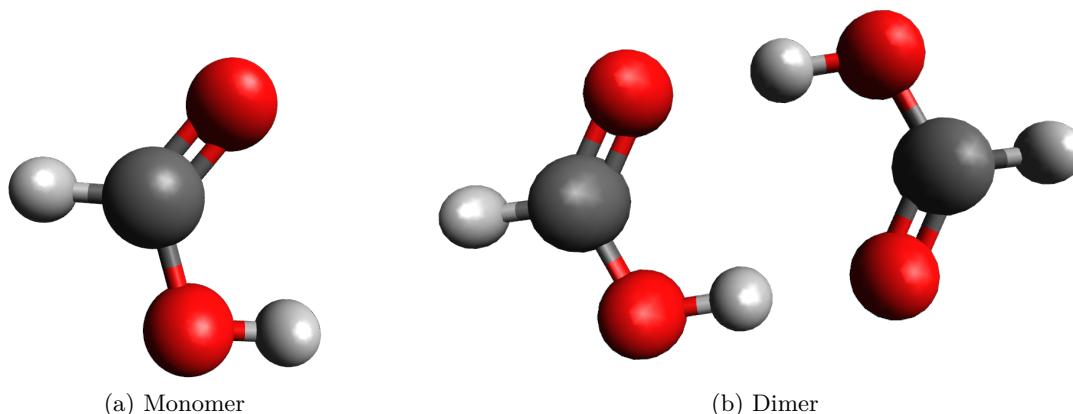


Figure 3: Monomer and dimer structures for formic acid. Geometries were optimized on the TPSS-D3/def2-TZVP level of theory.

The dimerization energy of formic acid is defined as the difference between the dimer energy and the sum of the monomer energies:

$$\Delta E_{\text{dimerization}} = E_{\text{dimer}} - 2 \cdot E_{\text{monomer}}, \quad (2)$$

where E_{dimer} is the energy of the dimer and E_{monomer} is the energy of the monomer.

The dimerization energy was computed for three different methods, namely HF, MP2^[4] and TPSS-D3, using the cc-pVXZ^[13] and aug-cc-pVXZ basis sets, where X = D, T, Q. The dimerization energy was computed for each method and basis set combination with optimized geometries at the TPSS-D3/def2-TZVP level. The results are shown in Figure 4.

The dimerization energy for formic acid shows a strong dependence on the choice of basis set and method. The aug-cc-pVXZ basis sets generally yield more negative dimerization

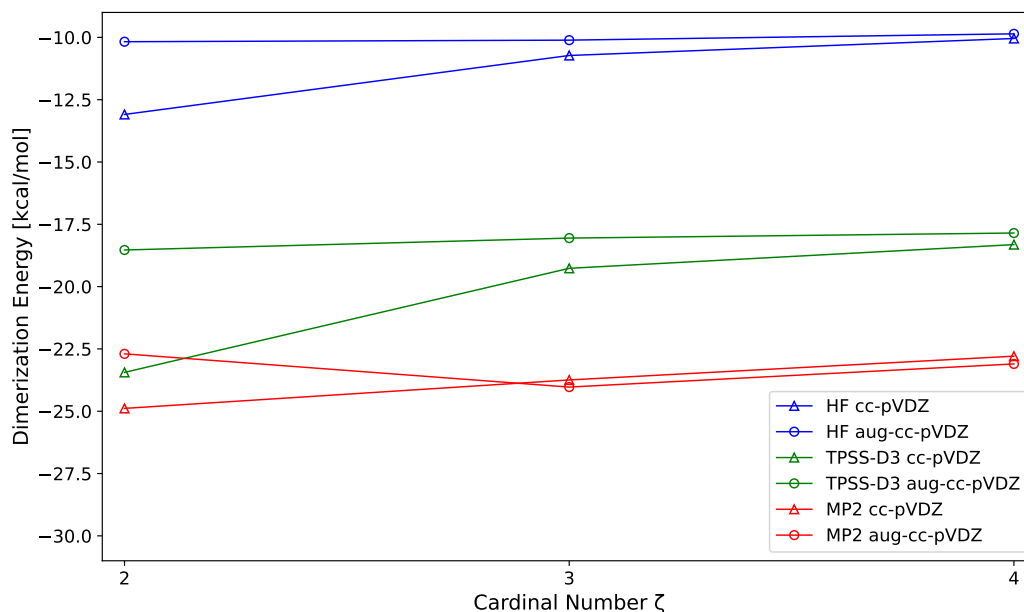


Figure 4: Convergence behaviour for the formic acid dimer aug-cc-pVXZ and cc-pVXZ basis sets at HF, MP2, and TPSS-D3 levels of theory.

energies compared to the cc-pVXZ basis sets. For a cardinal number of four, all methods and basis sets are at different dimerization energies, indicating that the choice of method has a significant impact on the computed dimerization energy.

The calculations generally contain two different errors, the basis set incompleteness error (BSIE) and the basis set superposition error (BSSE). The BSIE is the error that arises from the finite size of the basis set and is the difference between the complete basis set (CBS) limit and the energy computed with a finite basis set. This error is always present and not only limited to dimerization energies. The BSIE can be reduced by using larger basis sets and can also be fully eliminated by using non-Gaussian basis sets, which were not used in this report.

It is important to note that the convergence with respect to the basis set depends on the method used. For instance, SCF energies converge exponentially with respect to the basis set size, while correlation energies converge with inverse cubic behavior. HF energies converge faster than MP2 energies, which confirms the convergence behaviour, since HF is a SCF method and MP2 is a post-SCF method that accounts for electron correlation. TPSS-D3 is a meta-GGA functional, which also accounts for correlation, and the dimerization energy shows a convergence behaviour that lies between HF and MP2.

HF and TPSS with the cc-pVXZ basis sets show very similar convergence behaviour. Both exhibit almost no change in the dimerization energy, indicating that the dimerization energy is already converged with respect to the basis set size. The MP2 method with the cc-pVXZ basis sets shows a larger change in the dimerization energy, indicating that the dimerization energy is not yet converged with respect to the basis set size.

The aug-cc-PVXZ basis sets show a different convergence behaviour for the three methods. The HF and TPSS methods again show a similar convergence behaviour. Both methods exhibit a more pronounced change in the dimerization energy with increasing basis set size, indicating that the dimerization energy is not yet fully converged. The MP2 method with the aug-cc-PVXZ basis sets shows a constant change in the dimerization energy, indicating that the dimerization energy is not yet converged with respect to the basis set size.

The BSSE is the error that arises from the interaction of the basis functions of the two monomers in the dimer. This error is present in energies for the dimer or other intramolecular interactions. The BSSE is generally larger for an increased basis sets, because the interaction between the basis functions of the two monomers is stronger. The BSSE compensates the BSIE and leads to a artificially lowered dimerization energy. The BSSE is only not present if one uses a minimal basis set, which is not the case here.

Both errors can be reduced by using larger basis sets, but it is not possible to eliminate them completely, only by theoretically using a CBS. Another possibility is the use of Boys-Bernardi counterpoise correction, which is a method to correct for the BSSE.

The total energies of the formic acid monomer and dimer for the different methods and basis sets are shown in Figure 5. The total energies were computed using the same method and basis set as the dimerization energies.

The total energies in Figure 5 show a similar convergence behaviour for the monomer and dimer for each method and basis set type. Again, HF and TPSS-D3 show similar convergence behaviour, while MP2 shows a different convergence behaviour, as already explained above.

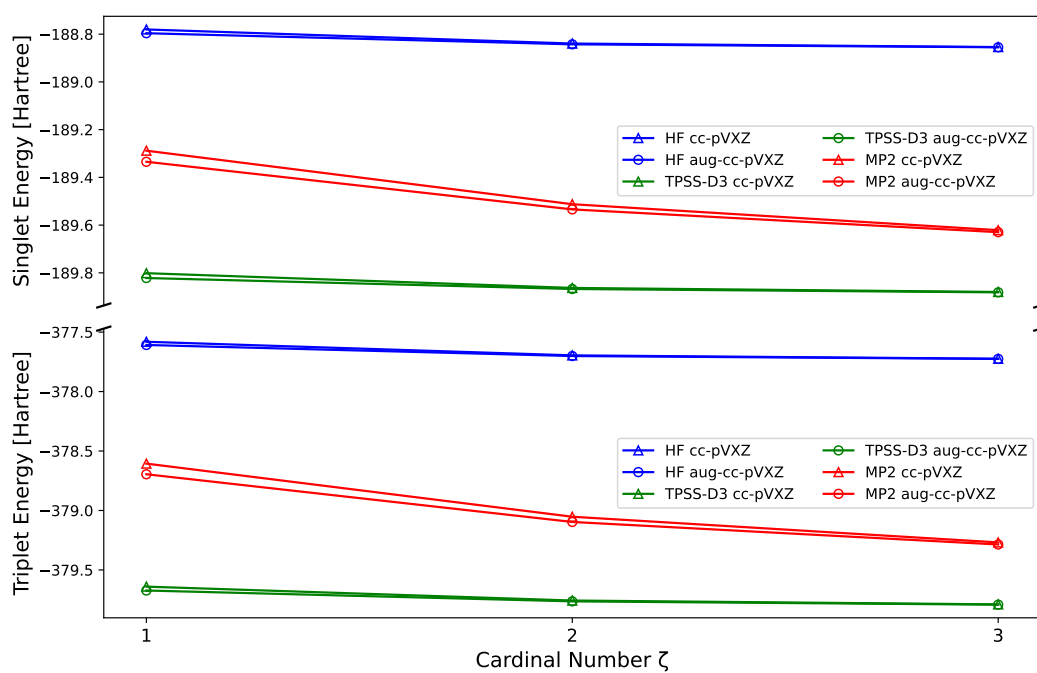
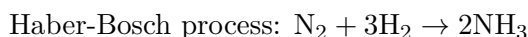


Figure 5: Total energies for the formic acid monomer and the formic acid dimer with aug-cc-pVXZ and cc-pVXZ basis sets at HF, MP2, and TPSS-D3 levels of theory.

3 Thermochemistry

In quantum chemistry, thermodynamics provides an understanding of energy changes at the molecular level in industrial processes such as the Haber–Bosch synthesis of ammonia and the steam reforming of methane. This enables accurate predictions of reaction enthalpies, equilibria, and efficiencies from first principles. For this reason, the reaction enthalpies ΔH for the Haber-Bosch process and the steam-reforming of methane were computed using the different level of theory.



The geometries were optimized on the TPSS^[8]-D3^[12]/def2-TZVP^[1] level of theory using TURBOMOLE V7.7.1.^[11] In addition, the geometries of the molecules involved in the Haber Bosch process were also optimized at the MP2^[4]/def2-TZVP level of theory and the RMSD to the TPSS-D3/def2-TZVP geometries was computed using the software RMSD.^[14] The results are shown in Table 2. The RMSD values are in the order of 10^{-3} Å and show that the geometries are very similar. This indicates that the choice of the method does not have a large impact on the geometry and the selection of the method can mainly be based on the computational cost. Furthermore, this shows that the TPSS-D3/def2-TZVP level of theory is good enough for the thermal corrections, as the change between TPSS-D3 and MP2 is negligible.

Table 2: RMSD between TPSS-D3 and MP2 geometries optimized in a def2-TZVP basis for all molecules involved in the Haber-Bosch process.

	RMSD (Å)
N ₂	$5.7 \cdot 10^{-3}$
H ₂	$3.1 \cdot 10^{-3}$
NH ₃	$6.6 \cdot 10^{-3}$

The reaction enthalpies of both reactions were computed using the MP2 and B3LYP^[6,7]-D3^[12] methods in a def2-TZVP basis set and the Becke 2-parameter exchange functional with Perdew-Lee-Yang-Parr correlation functional (B2PLYP)^[15]-D3^[12] and CCSD(T) in a def2-QZVP^[1] basis set. First, the single point energies were computed using the optimized geometries from the TPSS-D3/def2-TZVP level of theory. The single point

energies were then corrected for the thermal corrections at 298 K, which were calculated at the TPSS-D3/def2-TZVP level of theory. The results are shown in Table 3.

Table 3: Reaction enthalpies in kcal/mol for the Haber-Bosch process and steam-reforming of methane. Geometries were optimized on the TPSS-D3/def2-TZVP level. B2PLYP-D3 and CCSD(T) energies were computed in a def2-QZVP basis, def2-TZVP was used for all other methods.

Reaction	TPSS-D3	MP2	B3LYP-D3	B2PLYP-D3	CCSD(T)
Haber-Bosch	-12.7	-12.1	-21.0	-23.9	-24.5
	Exp: ^[10] -22.5				
Steam-Reforming	-42.4	43.1	47.3	50.4	51.4
	Exp: ^[10] 49.3				

In total, all methods yield a reasonable reaction enthalpies. The MP2 and the B3LYP-D3 methods underestimate the reaction enthalpies for both reactions, while the B2PLYP-D3 and CCSD(T) methods overestimate them.

The B2PLYP-D3 method yields the best agreement with the experimental values for both reactions, while the CCSD(T) method is slightly worse. The MP2, TPSS-D3 and B3LYP-D3 methods are not as accurate.

The general trend of the *Jacob’s Ladder* is observed for this example, where the more accurate methods yield better results. The hybrid functional B3LYP-D3 is more accurate than the MP2 method and the meta-GGA TPSS-D3, but not as accurate as the double hybrid B2PLYP-D3. Furthermore, the CCSD(T) and the B2PLYP-D3 methods used a larger basis set than the MP2 and B3LYP-D3 methods, which also leads to more accurate results. It is expected that the CCSD(T) method yields more accurate results than the B2PLYP-D3 method, but this is not the case here. This could be due to the fact that the CCSD(T) method converges more slowly with respect to the basis set than the B2PLYP-D3 method, which is why the B2PLYP-D3 method yields better results in this case. The CCSD(T) method is also more expensive than the B2PLYP-D3 method.

4 Kinetics

The kinetic isotope effect describes the change in reaction rate that occurs when an atom in the reactants is replaced by one of its isotopes, most commonly hydrogen by deuterium. This phenomenon provides valuable insight into reaction mechanisms, as differences in mass affect vibrational frequencies and zero-point vibrational energy (ZPVE), thereby influencing activation barriers and rate constants.

The kinetic isotope effect for the reaction of methane with a hydroxyl radical was studied using the B3LYP^[6,7]-D3^[12]/def2-TZVP^[1] level of theory for a transition state search, which was performed using TURBOMOLE V7.7.1.^[11]

The transition state research was done for $\text{CH}_4 - \text{OH}$ and $\text{CD}_4 - \text{OH}$, where the latter contains the deuteriated isotopologue of methane. The geometry of the transition state is shown in Figure 6.

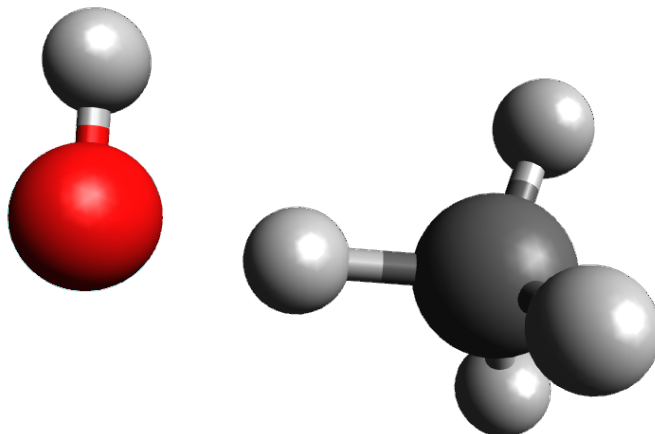


Figure 6: Geometry of the transition state for the reaction of methane with a hydroxy radical. The transition state was optimized on the B3LYP-D3/def2-TZVP level of theory.

The electronic energies, ZPVE and thermal corrections at 298 K for the reactants and transition states are shown in Table 4. The electronic energies were computed using the B3LYP-D3/def2-TZVP level of theory. The ZPVE and thermal corrections were computed using the same level of theory.

The energies in Table 4 amount to

Table 4: Electronic energies (E_h) along with zero-point vibrational energies and thermal corrections (298 K) for reactants and transition states.

	CH ₄	CD ₄	OH	CH ₄ - OH	CD ₄ - OH
E_{el} [kcal/mol]	-25.415	-25.415	-47.524	-72.937	-72.937
ZPVE (kcal/mol)	27.959	20.500	5.256	31.749	25.217
Thermal correction (kcal/mol)	2.393	2.468	2.074	3.475	3.732

$$\Delta H_H^\ddagger = -0.67 \text{ kcal/mol}$$

$$\Delta H_D^\ddagger = +0.44 \text{ kcal/mol}$$

with the subscripts H and D referring to regular and deuteriated methane respectively.

As seen in Table 4, the transition state for the reaction of methane with a hydroxy radical is lower in energy than the transition state for the reaction of deuteriated methane with a hydroxy radical due to the fact that the ZPVE and the thermal corrections are lower for the regular methane. The electronic energies are the same for both transition states, because they do not depend on the mass of the nuclei in the molecules.

The thermal corrections in contrast depend on the mass of the nuclei, which is why the thermal correction for the transition state of deuteriated methane is larger than for regular methane. The ZPVE is given by:

$$\text{ZPVE} = \frac{1}{2}h\nu, \quad (3)$$

where h is the Planck constant and ν are the vibrational frequencies of the system. The vibrational frequencies, which are taken in account for the thermal correction, can be computed from the harmonic oscillator model:

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}, \quad (4)$$

where k is the force constant and μ is the reduced mass of the system. The reduced mass can be computed from the masses of the two atoms involved in the reaction:

$$\mu = \frac{m_1 \cdot m_2}{m_1 + m_2}. \quad (5)$$

The reduced mass for the transition state of regular methane is smaller than for the transition state of deuteriated methane, which leads to larger vibrational frequencies and therefore a smaller thermal correction.

The relative reaction rates can then be determined from $\Delta H^\ddagger$ according to:

$$\frac{k_H}{k_D} = \exp\left(\frac{\Delta H_D^\ddagger - \Delta H_H^\ddagger}{RT}\right) = 6.52 \quad (6)$$

This indicates that the reaction of methane with a hydroxy radical is about 6.5 times faster than the reaction of deuteriated methane with a hydroxy radical.

Wiberg et al.^[16] investigated the deuterium isotope effect for different reactions, not including the reaction of methane with a hydroxy radical, and found relative reaction rates $\frac{k_H}{k_D}$ between 1.2 and 10. As the exact same reaction was not investigated, it is not possible to compare the results directly. However, the result of 6.5 is in line with the results of Wiberg et al.,^[16] which indicates that the method used in this report is suitable for the calculation of kinetic isotope effects.

For example, the reaction of hydrogen with oxygen is about 1.56 times faster than the reaction of deuterium with oxygen at a temperature of 560 °C and the reaction of methyl radicals with hydrogen is about 5.5 times faster at 130 °C than the reaction of methyl radicals with deuterium.^[16] This indicates that the kinetic isotope effect is significant for reactions involving hydrogen and deuterium, which is also observed in this report, and furthermore, the results suggest that the method used here is reliable for predicting kinetic isotope effects.

5 Solvation

The S_N2 reaction of chloromethane with fluoride is a well-known reaction in organic chemistry. The reaction proceeds via a transition state, where the nucleophile attacks the electrophile and the leaving group is expelled.

The solvation effects on the reaction can be significant, as the presence of a solvent can stabilize certain states and alter the reaction pathway. In this case, we consider the solvation effects of methanol on the S_N2 reaction of chloromethane with fluoride. The geometry of the transition state in vacuum is shown in Figure 7.

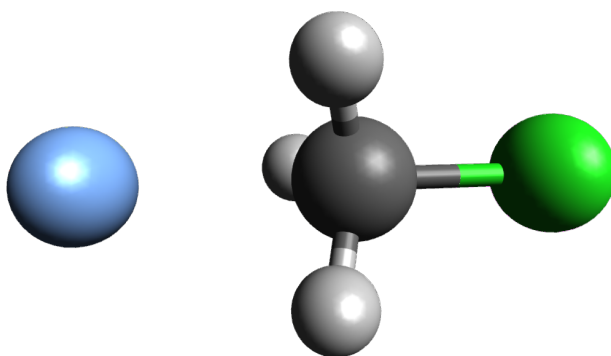


Figure 7: Geometry of the transition state for the S_N2 reaction of Chloromethane and Fluoride. The transition state was optimized on the PW6B95-D3/def2-TZVP level of theory. The C-F distance in the figure is 4.00 Å in vacuum.

The potential energy curves for the S_N2 reaction of chloromethane with fluoride in vacuum and in the presence of methanol as an implicit solvent were computed using the PW6B95^[9]-D3^[12]/def2-TZVP^[1] level of theory. Both calculations were performed using TURBOMOLE V7.7.1.^[11] The potential energy curves were computed for C-F distances between 2.25 Å and 8.0 Å in vacuum and in the presence of methanol as an implicit solvent using the COSMO^[17] model. The potential energy curves are shown in Figure 8.

The potential energy curves in Figure 8 show that the reaction proceeds via a transition state, where the C-F distance is about 3.75 Å in vacuum and about 4.00 Å in the presence of methanol as an implicit solvent. The transition state is higher in energy in the presence of methanol. The different C-F distances in vacuum and in the presence of methanol are due to the charge screening effect of the solvent.

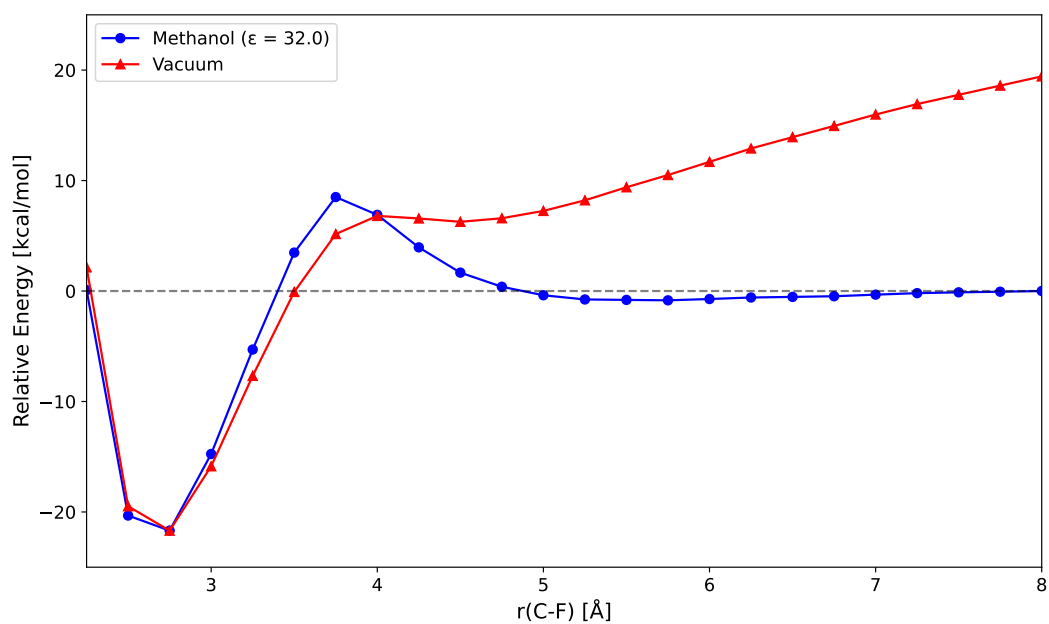


Figure 8: Relative energy curve for the $\text{S}_{\text{N}}2$ reaction of Chloromethane and Fluoride. Blue circles: In presence of methanol as an implicit solvent. Red triangles: In vacuum. The blue curve has been adjusted to the dissociation limit and the red curve was shifted such that both minima align.

The reaction barriers for the reactions can be estimated based on the potential curves:

$$\Delta E_{\text{solv}}^{\ddagger} = 9.3 \text{ kcal/mol}$$

$$\Delta E_{\text{gas}}^{\ddagger} = 0.5 \text{ kcal/mol} .$$

The reaction barrier from large C-F distances to smaller C-F distances in the presence of methanol as an implicit solvent is significantly higher than in vacuum, which indicates that the solvation effects are significant. In the other direction, the reaction barriers are very similar. This indicates that the COSMO model has the largest effect for large C-F distances, where the solvation effects for the fluoride ion are more pronounced.

Nevertheless, the reaction barrier can not be compared this easy, because the fluoride ion in vacuum does not experience the electrostatic interactions with the solvent. The COSMO model stabilizes the incoming fluoride ion, the leaving chloride ion and the transition state, which leads to a higher reaction barrier in the presence of methanol. The reaction barrier in vacuum is lower, because the fluoride ion is not stabilized by the charge screening and can therefore react more easily with chloromethane. When the fluoride ion approaches chloromethane, it experiences a stronger electrostatic interaction with the chloromethane, which leads to a stabilization of the fluoride ion. The negative charged fluoride ions are stabilized by the positive partial charges of the chloromethane molecule.

The behavior for large C-F distances in vacuum are observed due to the fact that the fluoride ion is a highly charged small anion without any stabilization of the charge. The charge screening of the solvent for the reaction in methanol stabilizes the fluoride ion and shows a more realistic behavior for large C-F distances.

6 Activation Energies

The reaction path for the Claisen rearrangement of allyl-vinyl ether and the Diels-Alder reaction of cyclopentadiene to *endo*-dicyclopentadiene were computed using the double ended growing string method (GSM) at the GFN2-xTB^[18] level. First, geometries of the reactant and product were optimized using the GFN2-xTB method. Then, the double ended GSM was used to compute the reaction path between the reactant and product of both reactions for 20 points. The reaction paths are shown in Figure 9 and Figure 10.

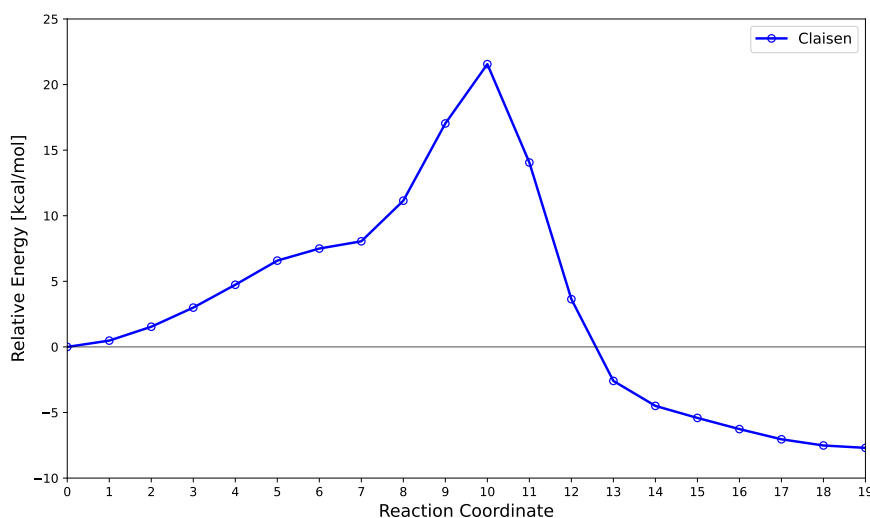


Figure 9: Relative energy along the reaction coordinate for the Claisen rearrangement of allyl-vinyl ether.

The activation energy for the Claisen rearrangement of allyl-vinyl ether is about 21.6 kcal/mol. The reaction proceeds via a cyclic transition state, where the C-O bond is broken and the C-C bond is formed. The energy of the reactants is lower in energy than the products, which indicates that the reaction is exothermic.

This activation energy is smaller than the experimental value of 30.6 kcal/mol,^[19] which suggests that the computational method used may underestimate the activation energy. This indicates that the transition state is not described accurately enough, which can be due to the limitations of the GFN2-xTB method. The GFN2-xTB method is a semi-empirical method that is not as accurate as ab initio methods, but it is computationally efficient and can be used for larger systems.

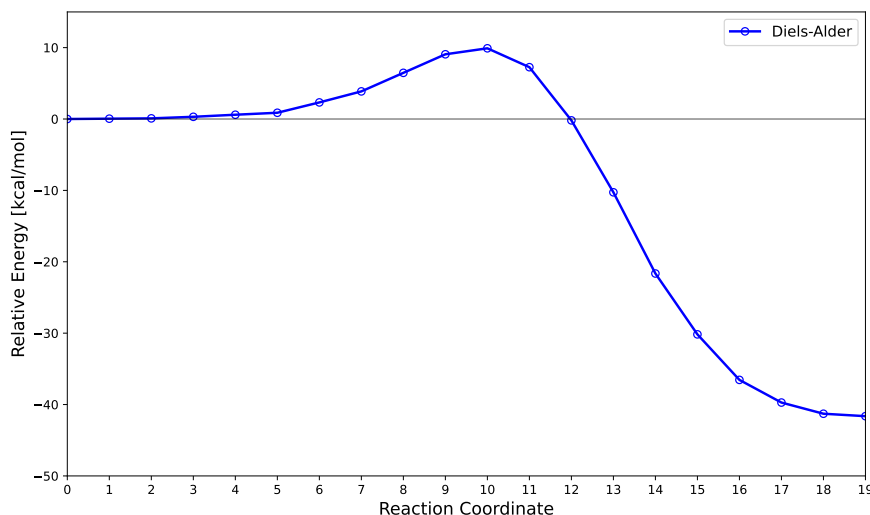


Figure 10: Relative energy along the reaction coordinate for the Diels-Alder reaction of cyclopentadiene to *endo*-dicyclopentadiene.

The activation energy for the Diels-Alder reaction of cyclopentadiene to *endo*-dicyclopentadiene is about 9.9 kcal/mol. The reaction proceeds via a transition state, where the C-C bond is formed and the C=C bond is broken.

The GFN2-xTB method is a semi-empirical method that is not as accurate as ab initio methods, but it is computationally efficient and can be used for larger systems. It does not account for anharmonic vibrational corrections, which can lead to an underestimation of the activation energy. In addition, the GFN2-xTB method does not account for high-level electron correlation. It is parameterized against higher-level data, but it is still a approximation.

The reliability of the activation energy for both reactions can be improved by using a higher level DFT method. This would provide a more accurate description of the reaction path, but would also increase the computational cost. Further improvements can be made by more advanced optimizing of the input geometries of the reactants and products, as this has a large impact on the used method.

The implementation of these two improvements was precluded by the tutor's instructions.

7 Noncovalent Interactions

Noncovalent interactions are important in many chemical and biological processes. In this section, the interaction of methane with argon and krypton is investigated. One representative geometry of the methane-argon complex is shown in Figure 11.

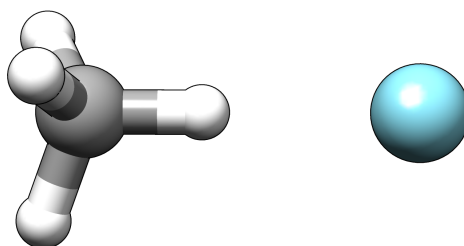


Figure 11: Geometry of the $\text{CH}_4 \cdots \text{Ar}$ complex. Figure taken from reference [10].

The potential energy curves for the interaction of methane with argon and krypton were computed using the following methods: Becke-Lee-Yang-Parr (BLYP),^[7,20] BLYP^[7,20]-D3^[12] and MP2.^[4] All calculations were performed using the def2-QZVP^[1] basis set. The potential energy curves were computed for $\text{CH}_4\text{-Ar/Kr}$ distances between 4.5 Å and 15.0 Å, were normalized to the dissociation limit and are shown in Figure 12.

The potential energy curves in Figure 12 show the importance of dispersion interactions for noncovalent interactions. The BLYP method does not account for dispersion interactions and therefore yields a repulsive potential energy curve, which is not physically meaningful. The dispersion interactions are not included in the BLYP method, because it is a GGA functional that only accounts for the exchange and correlation energy of the electrons, which only uses the local electron density and its gradient. In contrast, dispersion interactions are non-local and require a different treatment. The BLYP method is therefore not suitable for describing noncovalent interactions.

The BLYP-D3(BJ) method is the same GGA functional that does not account for dispersion interactions, but the D3 dispersion correction is added to the energy expression. This yields a potential energy curve that is in good agreement with the MP2 method.

The MP2 method yields a potential energy curve that is slightly lower in energy than the BLYP-D3(BJ) method. The MP2 method also includes dispersion interactions explicitly.

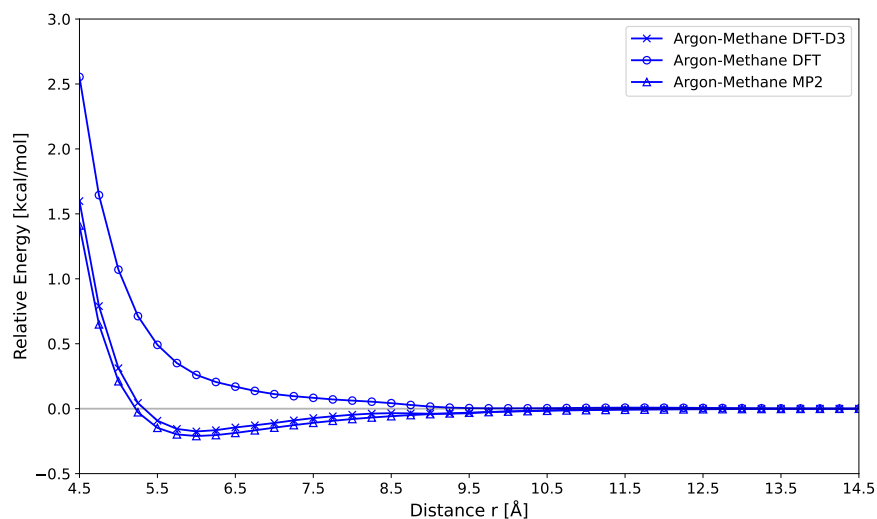
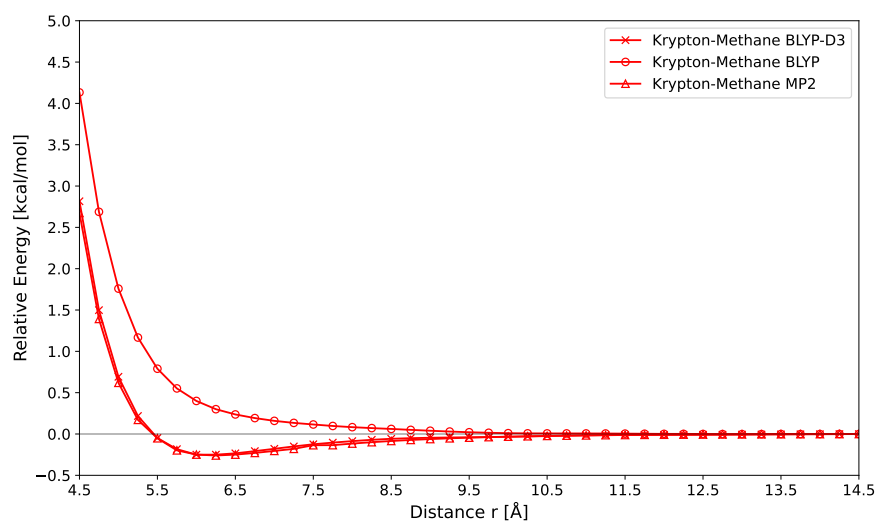
(a) $\text{CH}_4 \cdots \text{Ar}$ (b) $\text{CH}_4 \cdots \text{Kr}$

Figure 12: Potential energy curves for methane with argon (a) and krypton (b).

Both methods yield a minimum at around 6.0 Å for the CH₄-Ar complex and for the CH₄-Kr complex.

Argon and krypton are noble gases and do not form covalent bonds with methane. Argon is the lighter of the two and therefore has a smaller van der Waals radius, which leads to a lower polarization and a weaker dispersion interaction. This results in a slightly lower binding energy for the CH₄-Ar complex compared to the CH₄-Kr complex.

The only methods that should really be considered for these types of interactions are those that include dispersion corrections, such as MP2 and BLYP-D3(BJ), where the latter inherently includes dispersion interactions and has a slightly higher accuracy. On the other hand, the MP2 method is more computationally expensive than the BLYP-D3(BJ) method and therefore not always feasible for larger systems.

8 Spectroscopy

8.1 IR-Spectroscopy

The IR-spectra of 1,4-Benzoquinone were computed on the TPSS^[8]-D3^[12]/def2-SVP^[1] and HF-D3^[12]/def2-SVP levels of theory using TURBOMOLE V7.7.1.^[11] The structure of 1,4-Benzoquinone is shown in Figure 13.

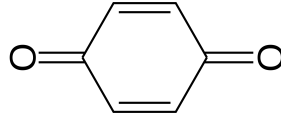


Figure 13: Structure of 1,4-Benzoquinone.

First, the geometry of 1,4-Benzoquinone was optimized with TURBOMOLE V7.7.1^[11] using the respective level of theory. Then, the vibrational modes were computed using the subprogram aoforce. The resulting IR-modes are shown in Table 5. Furthermore, the scaling factor for the IR-modes was calculated using the experimental values from [10] and the theoretical values from the TPSS-D3/def2-SVP and HF-D3/def2-SVP levels of theory. The scaling factor was calculated using the following equation:

$$f_{\text{scal}} = \frac{\tilde{\nu}_{\text{exp}}}{\tilde{\nu}_{\text{theo}}}, \quad (7)$$

where $\tilde{\nu}_{\text{exp}}$ is the experimental wavenumber and $\tilde{\nu}_{\text{theo}}$ is the theoretical wavenumber.

Table 5: Wavenumbers of IR-modes (cm^{-1}) calculated at the HF-D3/def2-SVP and TPSS-D3/def2-SVP level along an experimental reference [10]. Each theoretical frequency has been assigned a scaling factor (see Eq. (7))

Exp ^[10]		HF-D3			TPSS-D3		
$\tilde{\nu}$	I	$\tilde{\nu}$	f_{scal}	I_{rel}	$\tilde{\nu}$	f_{scal}	I_{rel}
3075.0	0.01	3387.6	0.91	0.01	3133.74	0.98	0.01
1670.0	1	2028.65	0.82	1	1704.54	0.98	1
1290.0	0.23	1452.21	0.89	0.17	1304.92	0.99	0.29
1070.0	0.15	1167.24	0.92	0.06	1036.34	1.03	0.15
870	0.3	988.47	0.88	0.17	885.49	0.98	0.23

The averaged scaling factors for the IR-modes were found to be 0.88 for HF-D3 and 0.99 for TPSS-D3. This indicates that the TPSS-D3 method yields more unscaled results than

the HF-D3 method for the IR-spectra of 1,4-Benzoquinone. Both methods overestimate the wavenumbers, but the TPSS-D3 method is almost identical to the experimental values. This large overestimation for HF-D3 is due to the underestimation of the bond length. TPSS-D3 is a meta-GGA that includes the kinetic energy of the electrons inside the exchange correlation potential, which leads to a better description of the bond strength and therefore a better description of the IR-spectra. Nevertheless, the electron correlation included in TPSS-D3 is the main reason for the better agreement with the experimental values.

The calculations could be improved by using a larger basis set or using a higher level of theory. However, this would increase the computational cost significantly. It is therefore important to find a balance between accuracy and computational efficiency when performing such calculations, which leads to the choice of the TPSS-D3/def2-SVP level of theory for this task, as it is accurate enough.

8.2 The Color of Indigo

The color of indigo is a well-known example of a molecule that exhibits strong color due to its electronic structure. The structure of indigo was optimized using the TPSS^[8]-D3^[12]/def2-SVP^[1] level of theory and is shown in Figure 14.

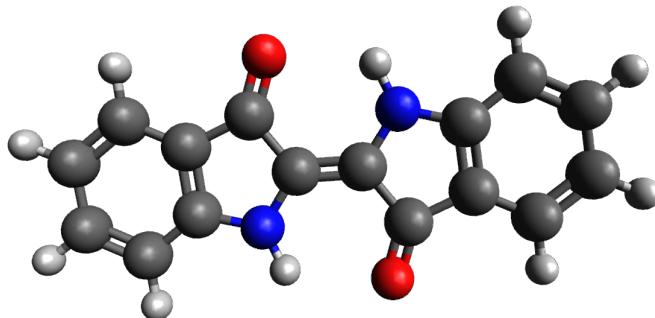


Figure 14: Optimized geometry of indigo on the TPSS-D3/def2-SVP level of theory.

The optical transitions of indigo were calculated using time-dependent HF method and time-dependent DFT methods with the Perdew-Burke-Ernzerhof (PBE)^[21] and Perdew-Burke-Ernzerhof with 25% Hartree-Fock exchange (PBE0)^[22] functionals in a def2-SVP basis set. The results are shown in Table 6.

Table 6: Optical transition of the indigo molecule calculated using different methods in the def2-SVP basis set.

	HF	PBE	PBE0	Exp ^[10]
λ (nm)	373.9	618.7	533.5	610
Colour	yellow/orange	blue	violet/red	blue (solution/solid)

The PBE and PBE0 methods yield a blue and violet color, respectively, while HF yields a yellow/orange color. The experimental value is in good agreement with the PBE method.

The PBE0 method is not in agreement with the experimental value. The HF method also does not match the experimental values and has even more significant deviations than PBE0. This indicates that the HF method is not suitable for the calculation of optical transitions of indigo at all, because it does not account for electron correlation, which is important for the description of the electronic structure of indigo and the excitation energies.

The PBE method is a GGA functional that should perform worse than the hybrid PBE0 functional according to *Jacob's Ladder*, but in this case it yields a better result. This is due to the fact that the PBE0 functional is based on the PBE functional, but includes 25 % of the exact HF exchange energy. This leads to a larger deviation from the experimental value.

8.3 NMR Parameters

^{13}C NMR Shifts for Organic Compounds

The ^{13}C NMR shifts for several organic compounds were calculated using the Perdew-Burke-Ernzerhof hybrid with 3 corrections (PBEh-3c)^[23] method for geometry optimization and the PBE^[21]/pcSeg-2^[24] method for the calculation of the shielding tensors in ORCA 5.0.4.^[2,3] PBEh-3c is based on PBE and includes exact HF exchange as it is a hybrid (h). It further includes three corrections (3c): dispersion correction, BSSE correction and geometrical counterpoise correction. The Lewis structures of the compounds are shown in Figure 15.

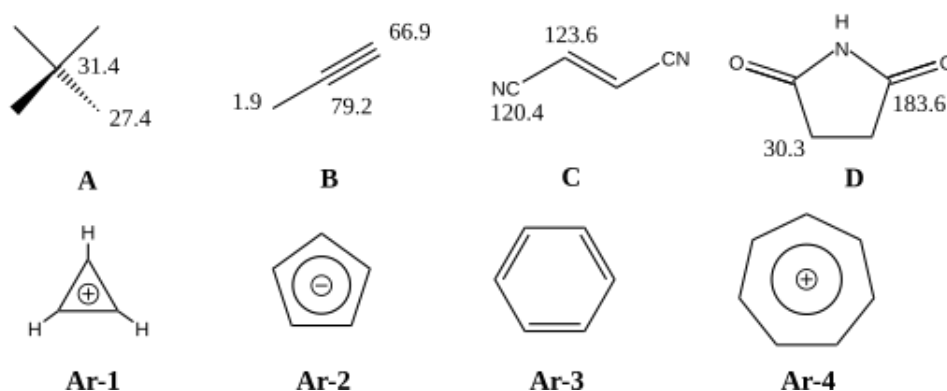


Figure 15: Lewis structures of four organic compounds A – D with their experimentally obtained chemical shift as well as four aromatic compounds Ar-1 - Ar-4. Figure taken from reference [10].

The chemical shifts were calculated relative to tetramethylsilane (TMS) and the results are shown in Table 7.

The mean deviation (MD) and mean absolute deviation (MAD) values for the chemical shifts of the organic compounds are:

$$\text{MD} = \frac{1}{N} \sum_{i=1}^N (\delta_{\text{calc}} - \delta_{\text{exp}}) = 4.00 \text{ ppm}$$

$$\text{MAD} = \frac{1}{N} \sum_{i=1}^N |\delta_{\text{calc}} - \delta_{\text{exp}}| = 4.33 \text{ ppm.}$$

Table 7: ^{13}C chemical shifts (δ) for several organic molecules in reference to TMS. Geometries were optimized using PBEh-3c and shielding tensors were calculated at the PBE/pcSeg-2 level of theory.

Compound		δ (ppm)		
A	Calc	34.5	34.0	
	Exp	27.4	31.4	
B	Calc	4.03	71.2	85.4
	Exp	1.9	66.9	79.2
C	Calc	124.0	127.9	
	Exp	120.4	123.6	
D	Calc	33.9	182.3	
	Exp	30.3	183.6	

The similarity of the MD and MAD values indicates that the method has a systematic error in the chemical shifts, but the deviations are not too large. The largest deviation is for compound A, which has a difference of 7.1 ppm, and the smallest deviation is for compound D, which has a difference of -1.3 ppm.

The aromatic compounds Ar-1 - Ar-4 were also calculated using the PBE/pcSeg-2 method and their chemical shifts are shown in relation to their π -electron density ρ_π in Figure 16. The chemical shifts were calculated relative to TMS.

The chemical shifts of the aromatic compounds show a linear correlation with the π -electron density, which is expected due to the fact that the chemical shift is influenced by the electron density around the nucleus. The magnetic moment created by the aromatic ring current leads to an increased shielding of the nucleus, which results in a downfield shift of the chemical shift.

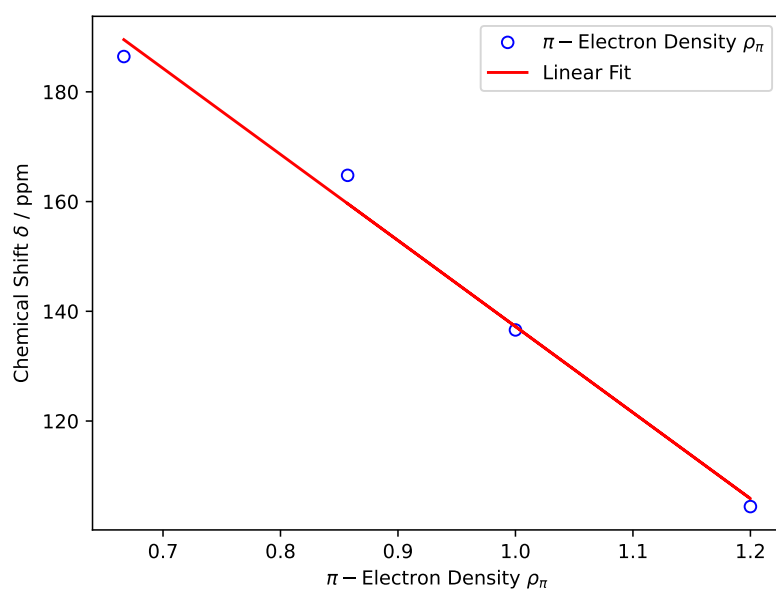


Figure 16: Chemical shifts δ of the aromatic compounds Ar-1 - Ar-4 plotted against the π -electron density ρ_π . The linear fit is shown as red line and the calculated values as blue circles. The chemical shifts were calculated using the PBE/pcSeg-2 level of theory.

^{13}C and ^{17}O NMR Shifts for Acetone

The effect of the solvent on the NMR shifts of acetone was investigated by calculating the chemical shieldings for acetone in gas-phase and in a solvent (water) using the PBE^[21]/pcSeg-^[24] method. The geometries were optimized in gas-phase and in a solvent using the COSMO^[17] model. The chemical shieldings were calculated relative to TMS for ^{13}C and relative to water for ^{17}O . The results are shown in Table 8.

Table 8: Chemical shieldings for acetone relative to TMS (^{13}C) and water (^{17}O) (δ in ppm). Reference was optimized and calculated in gas-phase.

Geometry	NMR	^{13}C		^{17}O	
		δ	$\Delta\delta_{\text{gas/gas}}$	δ	$\Delta\delta_{\text{gas/gas}}$
Gas	Gas	214.01	0.00	672.54	0.00
Gas	Solvent	224.08	10.07	616.54	-60.62
Solvent	Gas	216.17	2.16	682.76	10.23
Solvent	Solvent	226.25	12.24	618.83	-53.71
Experiment ^[10]		-		18.9	-75.5

The difference of the chemical shifts with the geometry in gas-phase and the NMR in solvent to the geometry in solvent and the NMR in gas-phase is shown in Table 8. The differences are 7.19 ppm for ^{13}C and 70.85 ppm for ^{17}O . This shows the importance of the choice of media for the NMR calculation.

In total, the chemical shifts for ^{13}C and ^{17}O in acetone calculated with the gas geometry and the solvent NMR matches the experimental value for ^{17}O , while the solvent geometry and solvent NMR matches the experimental value for ^{13}C .

The differences in the chemical shifts for gas-phase and solvent-phase for the NMR calculations are larger than that for the geometry. This indicates that the choice for the NMR calculation is more significant than the choice for the geometry, but it is still important to consider the solvent effects for both.

^1H NMR Shifts for Metacyclophane

The ^1H NMR shifts for in- and out-Metacyclophane were calculated using the PBEh-3c^[23] method for geometry optimization and the PBE^[21]/pcSeg-2^[24] method for the calculation of the shielding tensors in ORCA 5.0.4.^[2,3] The optimized geometries on the PBEh-3c method of in- and out-Metacyclophane are shown in Figure 17.

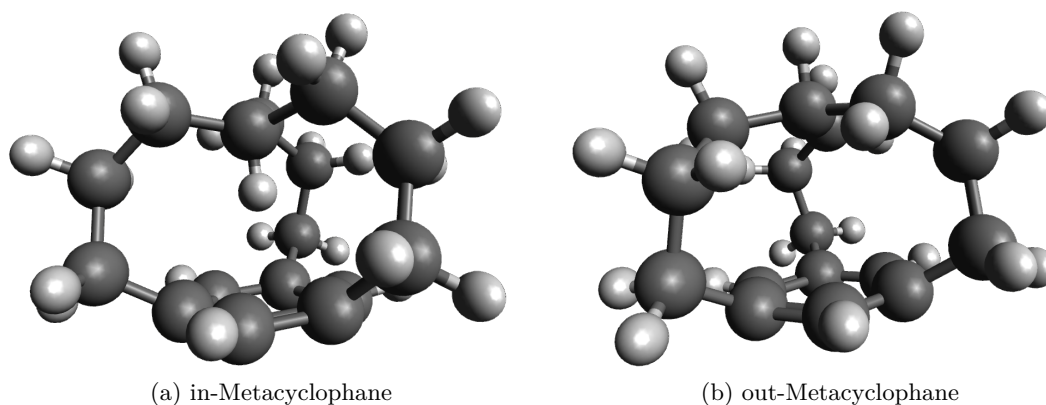


Figure 17: Optimized geometry of in- and out- $[3^{4,10}][7]$ Metacyclophane on the PBEh-3c level.

The chemical shifts were calculated relative to TMS and are:

$$\text{in: } \delta(^1\text{H}): -3.24$$

$$\text{out: } \delta(^1\text{H}): 2.07.$$

The chemical shift for in-Metacyclophane is more upfield, which indicates that the protons are stronger shielded by the aromatic ring current than for the out-Metacyclophane.

9 Appendix

Acronyms

B2PLYP Becke 2-parameter exchange functional with Perdew-Lee-Yang-Parr correlation functional

B3LYP Becke 3-parameter exchange functional with Lee-Yang-Parr correlation functional

BLYP Becke-Lee-Yang-Parr

BSIE basis set incompleteness error

BSSE basis set superposition error

CASPT2 complete active space with perturbation theory of second order

CASSCF complete active space self-consistent field

CBS complete basis set

CCSD(T) coupled cluster with single, double and perturbative triple excitations

DFT density functional theory

GSM growing string method

HF Hartree-Fock

MAD mean absolute deviation

MD mean deviation

MP2 Møller-Plesset perturbation theory of second order

PBE Perdew-Burke-Ernzerhof

PBE0 Perdew-Burke-Ernzerhof with 25% Hartree-Fock exchange

PBEh-3c Perdew-Burke-Ernzerhof hybrid with 3 corrections

PW6B95 Perdew-Wang 6-parameter exchange functional with Becke-95 correlation

RHF restricted Hartree-Fock

TMS tetramethylsilane

TPSS Tao-Perdew-Staroverov-Scuseria

UHF unrestricted Hartree-Fock

ZPVE zero-point vibrational energy

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